

03_Test_Plan

☐ 1. Introduction

The Test Plan defines the overall strategy for testing the OrangeHRM application using Selenium WebDriver and TestNG. It explains the scope, approach, resources, schedule, and deliverables of the automation testing process.

The main purpose of this test plan is to ensure that all critical functionalities of the OrangeHRM application are validated through structured and systematic test execution.

☐ 2. Test Objectives

The primary objectives of this test plan are:

- To validate login functionality with valid and invalid inputs
 - To verify forgot password functionality with different scenarios
 - To ensure navigation between pages works correctly
 - To validate UI messages and system responses
 - To ensure the automation framework runs without failures
 - To improve test coverage of critical application modules
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☐ 3. Scope of Testing

✓ In Scope

The following modules are covered in automation testing:

- Login Module
 - Valid login
 - Invalid login
 - Empty field validation
- Forgot Password Module
 - Valid username reset
 - Invalid username validation
 - Empty input validation
 - Cancel functionality
- Navigation Module
 - Dashboard navigation
 - Apply Leave page navigation
- UI Validation

- Error messages
 - Success messages
 - Page headings
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✗ Out of Scope

The following are not covered in this automation project:

- Performance testing
 - Security testing
 - Database testing
 - API testing
 - Mobile application testing
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□ 4. Test Strategy

The testing strategy used in this project is:

◆ Automation First Approach

All major functional test cases are automated using Selenium WebDriver.

◆ Framework Design

- Page Object Model (POM)
- TestNG framework for execution
- Java programming language

◆ Execution Strategy

- Tests are executed via testng.xml suite
- Browser execution supported for Chrome and Firefox

◆ Validation Strategy

- Assertions are used for verifying expected vs actual results
 - Error messages and UI elements are validated
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□ 5. Test Environment

◆ Hardware

- Standard PC or Laptop

◆ Software

- Windows OS
- Java JDK
- Eclipse IDE
- Selenium WebDriver
- TestNG Framework
- ChromeDriver / GeckoDriver

◆ Application

- OrangeHRM Demo Application
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□ 6. Test Data

◆ Login Data

- Valid Username: Admin
- Valid Password: admin123
- Invalid credentials for negative testing

◆ Forgot Password Data

- Valid Username: Admin
 - Invalid Username: random strings
 - Empty inputs
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□ 7. Test Execution Approach

The test execution process follows this flow:

1. TestNG suite (testng.xml) is executed
 2. BaseTest initializes browser and application URL
 3. Test classes are executed one by one
 4. Page classes perform actions on UI
 5. Assertions validate expected results
 6. Browser closes after execution
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☐ 8. Entry Criteria

Before starting test execution:

- Test cases must be designed and reviewed
 - Automation scripts must be ready
 - Browser drivers must be installed
 - Application must be accessible
 - Test data must be prepared
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☐ 9. Exit Criteria

Testing will be considered complete when:

- All test cases are executed
 - No critical defects remain open
 - Test execution reports are generated
 - Automation scripts run without configuration errors
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☐ 10. Deliverables

The following deliverables are produced:

- Automation test scripts
 - testng.xml execution suite
 - Test execution results
 - Bug reports (if any)
 - Screenshots of failures
 - Final test summary report
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☐ 11. Risks & Assumptions

⚠ Risks:

- UI changes in application may break locators
- Network issues may interrupt execution
- Browser compatibility issues

✓ Assumptions:

- Application remains stable during testing

- Test data is valid and available
 - Environment is properly configured
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□ 12. Summary

The test plan defines a structured approach for testing OrangeHRM application using automation. It ensures proper coverage of functional modules and provides a clear execution strategy for reliable and maintainable testing.
